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Non-invasive muscle-machine interface open source project: wearable hand myoelectrical orthosis (MES-FES)

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ABSTRACT

The paper describes the development of an open-source, low-cost, wearable hand myoelectrical orthosis (neuro-orthosis) device for people with hand disabilities. The device uses functional electrical stimulation (FES) driven by myoelectrical signals (MES) to assist hand movements, enabling users to perform daily activities with greater ease and independence. The device comprises a forearm-wearable device developed using the 3D additive manufacturing principle, allowing user customization. Fixed non-disposable electrodes are attached to the myoelectrical orthosis, aiding the correct positioning for the user. The whole control system is stand-alone, and parameters can be controlled by Bluetooth communication, making the device wireless. The paper describes the MES-FES device's design, development, and testing, including its technical specifications, usability, and effectiveness. The open-source project aims to provide an accessible and affordable solution for people with spinal cord lesions while contributing to the growing research on noninvasive muscle-machine interfaces.

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KEYWORDS

Do-It-Yourself; electromyography; functional electrical stimulation; neuro-orthosis; neuroprosthesis

Introduction

According to the World Health Organization, approximately 15% of the world's population has some disability (World-Health-Organization, The-World-Bank, on Disability WR, 2011). Developing countries like Brazil present a scenario where 8.9% of the total population has some disability, with 1.4% specifically facing difficulties in manual tasks, such as handling small objects or opening and closing containers (dos-Direitos-Humanos-e-da-Cidadania M, 2023). In recent years, noninvasive muscle-machine interfaces (MMIs) have gained popularity in biomedical engineering due to their potential to improve the

quality of life of people with neuromuscular disorders. The development of wearable devices that allow for real-time control of muscles through myoelectric signals has shown promise in assisting with daily tasks. These devices can help individuals with limited hand mobility grasp and hold objects that would otherwise be challenging (Martins et al., 2023). However, the cost and availability of these devices have been significant barriers for many individuals who would benefit from their use.

A neuroprosthesis is a technology that restores neurological functions by interacting with the nervous system via implantable or noninvasive interfaces (Marquez-Chin & Popovic, 2020; Shokur et al., 2021). Neuromodulation techniques such as functional electrical stimulation (FES) can restore movement by activating the user's muscles via short electrical pulses on the peripheral nervous system, delivered through electrodes on the skin, generating muscle contractions (Lynch & Popovic, 2008).

FES wearable devices show promise for assisting daily tasks and hand therapy but require technological improvements such as portability and ease of use before widespread clinical or home-based use. Limitations include the time spent on calibration and donning/doffing (Cardoso et al., 2022).

Wearable sensors can be applied in assistive technologies for people with disabilities, capturing physiological signals and translating them into commands to control intelligent equipment (Y. L. Chen et al., 2001; Khoshmanesh et al., 2021; Marquez-Chin & Popovic, 2020). Electromyography (EMG) is valuable sensing feedback information for real-time functional electrical stimulation (FES) control, as myoelectric signals (MES) provide muscle contraction levels and can be obtained from weak or partially paralyzed muscles (Kilgore et al., 2008; Zhang & Ang, 2007; Y. Zhou et al., 2018).

Several studies have proposed MES-FES interfaces for patients with neurological disorders in rehabilitation (Marquez-Chin & Popovic, 2020). For instance, Yoo et al. designed a 3D-printed myoelectric hand orthosis with computer-aided software and Polylactide (PLA) filament to enhance grasping function in individuals with chronic cervical spinal cord injury (SCI)-related hand weakness (Yoo et al., 2019). Controlled by electromyographic signals, the orthosis demonstrated improved hand function in a study involving 10 subjects. However, the authors did not present open-source features for the proposed solution. On the other hand, Ryser et al. presented a real-time, stand-alone, wearable myoelectric

intention detection system for stroke survivors using the Myo armband (Ryser et al., 2017). The hardware can be integrated into a robotic hand orthosis for all-day robot-assisted home rehabilitation. However, the authors only present and detail (with open-source codes) the acquisition device without evaluating it with hand orthosis. Portnova et al., on the other hand, aimed to enhance the accessibility, customizability, and functionality of an open-source wrist-driven orthoses (WDO) using 3D-printing and incorporating feedback from both SCI users and Prosthetics and Orthotics (P&O) students to provide quantitative and qualitative feedback on the fabrication, fit, and function of the WDO (Portnova et al., 2018).

Selvaraj Mercyshalinie et al. introduced an interactive hand orthosis named NOHAS to aid post-stroke individuals. The system features tendon-driven finger actuation mechanisms powered by servo motors, allowing users to replicate normal gripping behavior through a mobile app (Selvaraj Mercyshalinie et al., 2023). The NOHAS demonstrated promising results in finger movement characterization, energy efficiency, noise quantification, and functional tasks. Similarly, Abdallah et al. proposed a novel electromyography (EMG)-controlled 3D-printed hand exoskeleton to assist stroke victims in improving their gripping abilities (Abdallah & Bouteraa, 2023). Utilizing computer-aided design software and PLA filament, the exoskeleton's 3D architecture was created. The results suggest that this wearable exoskeleton could enhance hand rehabilitation by accommodating a broader range of motion and improving dexterity based on users' motion intentions. However, both studies did not present a fully open-source project to replicate or extend the findings.

Objectives

This paper aims to present the MES-FES device's design, development, and testing. It provides detailed information on the device's technical specifications, usability, and effectiveness. Our primary goal is to offer an accessible and affordable solution for individuals with neurological motor disabilities while contributing to the expanding field of noninvasive MMIs. The device was projected based on the following design criteria:

- **Low-Cost:** The device is intended to be affordable, making it accessible to a broader population of individuals with hand disabilities;
- **DIY (Do-It-Yourself):** Using off-the-shelf components encourages users to build the device themselves, which may help to spread the technology.
- **Non-Invasive:** The device was designed to be noninvasive, which makes it more acceptable for the general population;
- **Wearable:** The size and weight of the circuits and the design of the device housing were thought to be light enough to be wearable, enabling their use in daily life;
- **Customizable:** The forearm-wearable orthosis was developed using 3D additive manufacturing, allowing user customization to suit individual needs;

- **User-Friendly Interface:** The orthosis set-up should be intuitive and user-friendly, facilitating easy operation and adjustment.

Therefore, we present an original open-source project: a low-cost, DIY, noninvasive, customizable, wearable hand myoelectrical orthosis for people with hand disabilities. The device combines myoelectrical signals (MES) and functional electrical stimulation (FES) through a noninvasive surface peripheral nervous system interface (sPNSI). By detecting movement intention in real-time using surface electromyography (sEMG) sensors, the device triggers electrical stimulation to assist hand movements, enhancing users' independence and ease in daily activities. The forearm-wearable orthosis, developed using 3D additive manufacturing, allows for customization, while the stand-alone control system with Bluetooth communication ensures wireless operation. This customizable, lightweight model, accompanied by an open-source software platform, is designed to improve usability and applicability with off-the-shelf components.

Justification of need

Several works are making steps forward in the usability of MES-FES interfaces. The designs of these systems usually prioritize ease of wearability and convenience, featuring wearable structure, low weight, wireless communication, and voluntary activation (Crepaldi et al., 2021; Jasuja et al., 2019; Jovićić et al., 2012; Jung et al., 2021; Wang, Bi, et al., 2017). These features may encourage users to engage with the technology. Even when devices are designed for rehabilitation purposes, there is a growing concern about the compatibility of these devices with smartphones and their wearability (Wang, Guo, et al., 2017). Specifically, there are concerns about the ability of these devices to communicate with smartphones and their suitability for use as wearable technology.

In addition, by making the entire project accessible as open access and embracing an open-source approach, we aim to disseminate the technology and facilitate wider adoption by interested parties, ultimately impacting the lives of those with neuromuscular disorders. This encourages the scientific and medical communities to explore the potential of open-source technology in advancing assistive devices.

Design

In this section, the bracelet, circuits, and control strategy for the proposed myoelectrical orthosis 1 are presented and summarized in Figure 1.

Bracelet model

The bracelet was developed using additive manufacturing techniques, as it can be printed by 3D techniques using fused filament fabrication (FFF), separated into two parts. The bracelet part was printed using Thermoplastic Polyurethane (TPU), which presents flexibility for better coupling with the

¹All data from this paper will be available at GitHub, after publication.

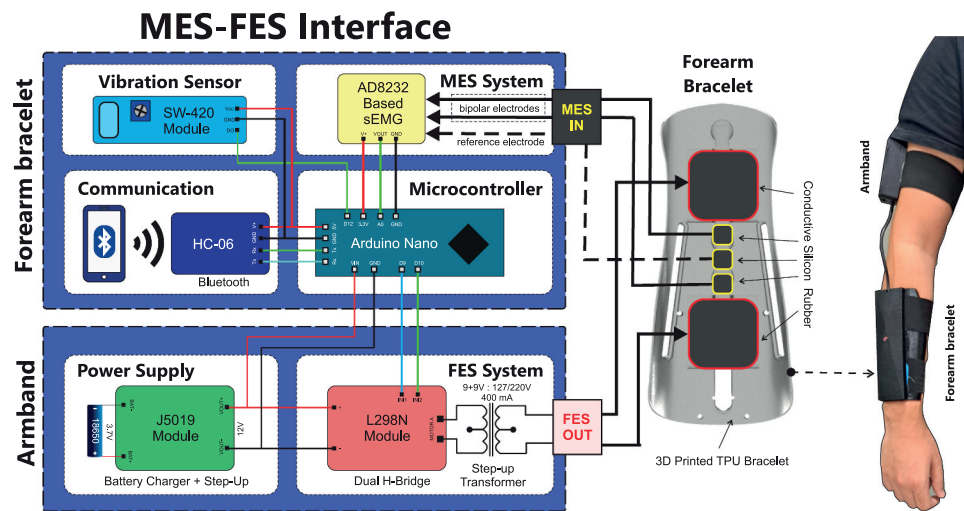


Figure 1. MES-FES interface and 3D-printed bracelet.

arm. The upper case part was printed with PLA, a cheap option, resulting in a rigid body for the piece. These parts are highlighted in Figure 1. The bracelet was modeled based on measures of the width and length of the subject forearm.

The electrodes used to measure the MES and apply the FES are mounted on the bracelet in a fixed position. This layout eases user positioning and reduces setup time. There are five electrodes in total: two for FES and three for MES, which are smaller than the FES electrodes.

The movement intention is detected using sEMG, which records the signal generated by the muscle (MES) during a contraction. An analog front-end acquires the MES to amplify the differential measure from a pair of electrodes. A third central electrode is used as a reference. The MES is converted to a digital signal using an Arduino Nano, a board based on the ATmega328P microcontroller. A script processes the MES to trigger the FES system, which modulates a pulse train to stimulate the surface peripheral nervous system and develop muscle contraction. A Bluetooth module (HC-06) is used to communicate the system with a mobile device to set parameters.

A vibration sensor based on the SW-420 sensor is installed inside the bracelet to avoid false positive detection caused by artifacts during non-contracting forearm movements. An external trimpot adjusts the vibration threshold. The microcontroller, the communication system, the vibration sensor, and the MES circuits are housed in the forearm bracelet.

A separate housing contains the power supply module, battery, and FES generating system. The J5019 module is a standalone linear Li-Ion battery charger (TP4056) coupled to a step-up converter. The system is powered by a single 18,650 battery (3.7 V) converted to 12 V. The pulse generated by the microcontroller is driven by an L298N module, which is a dual H-bridge DC driver. The driver output is amplified using an isolated step-up transformer. The module is mounted in an armband to reduce weight on the forearm, and it communicates with the forearm bracelet by shielded cables. The final FES output is connected to the external electrodes in the bracelet.

MES instrumentation

The MES circuit is based on a fully integrated single-lead biopotential acquisition front-end chip AD8232 (Devices, 2020). This integrated circuit features an instrumentation amplifier (with a fixed gain of 100), a 2-pole adjustable high-pass filter to attenuate electrode half-cell potential and motion artifacts, and a 3-pole low-pass filter with adjustable gain to tune the frequency band of interest for MES (20–500 Hz).

The circuit was set up to 850 (58.6 dB) gain and passing frequency band from

22.6 Hz to 503.3 Hz (Butterworth approximation), a typical CMRR of 80 dB, and a signal offset of 1.65 V. The MES is acquired by bipolar electrodes and a differential amplifier with high input impedance, which protects the user from electrical faults. A reference electrode establishes the signal baseline as half of the supply voltage. More details on the sEMG front-end can be found in (Mendes Junior et al., 2023).

Pilot evaluation with a participant with motor impairment

This study was conducted with the approval of the Ethical Committee of the State University of Londrina (protocol number 3.004.069).

Inclusion criteria

- Age over seven years (of both genders);
- Diagnosis of (i) spinal cord injury, (ii) brachial plexus injury, (iii) cerebral palsy, (iv) stroke, or (v) traumatic brain injury;
- Minimum of 8 months since the onset of the injury.

Exclusion criteria

- Health problems that hinder the continuity of the study;
- Intolerance to discomfort caused by electrical stimulation;

- Presence of neoplastic tissue near the electrically stimulated areas;
- Metallic implant in proximal regions or between electrodes of electrical stimulation;
- Cognitive or behavioral alterations that impede experimental intervention;
- Presence of other neurological or orthopedic disorders that prevent experimental intervention;
- Infectious contagious disease;
- Possession of bodily/vital maintenance implants that use electricity to function (such as a pacemaker);
- Wounds or other dermatological conditions that hinder the conduct of the study.

The data were obtained from a male participant (60 years old) with motor sequels of hemorrhagic stroke (06/17/2019 - lesion date) on the left body side.

MES processing

MES, acquired by A 12-bit ADC, was filtered using a second-order Butterworth digital bandpass of 10 to 40 Hz. This band was chosen to smooth the input and avoid the power line noise at 60 Hz. The raw signal was processed to detect the intention of movement to trigger the FES.

To improve the sensibility of the MES onset detection, an adapted linear threshold based on an adapted Taeger-Kaiser Energy Operator (TKEO) was applied on the rectified raw vector (Solnik et al., 2010). The MES onset is detected by comparing the signal to threshold (th) using the following relation:

$$MES_i > MES_{th} = (1 + th) \times MES_{i-1}, \quad (1)$$

in which MES_{i-1} and MES_i are the integrated mean absolute value (MAV) of two consecutive N-sample windows, which is the average of absolute MES value:

$$MES = \frac{1}{N} \sum_{k=1}^N |x_k|, \quad (2)$$

being x_k the k -th acquired data sample within the window. In this work, a window of 10 samples was chosen ($N=10$). Therefore, a threshold of $th=0$ means the detection of any increase in the MES value. Isolating the threshold term:

$$\frac{MES_i - MES_{i-1}}{MES_{i-1}} = \Delta MES > th, \quad (3)$$

in which ΔMES is the notation for a percentage difference in MES values. Thus, the threshold is a parameter that measures relative differences from one MES value to the next. For example, setting it at $th=0.1$ will detect when MES positively changes 10% about the previous epoch.

The difficulty in activating the FES driver is set by the threshold value th and should be individually tuned. This value represents the slope threshold between two consecutive MES windows. In this work, this value was adjusted at $th=0.16$. A point of difficulty in the app increases 0.01 in the threshold.

The general scheme of MES processing to trigger FES implemented on the micro-controller is presented in the flowchart in Figure 2.

To initiate the routine on Figure 2, it is necessary to calibrate the signals for the physiological range of the user. The procedure is executed by the user performing the maximal voluntary contraction (MVC) in the forearm muscles during wrist and hand extension. The obtained value is MES_{max} on Figure 2. Therefore, an anomalous signal peak over MES_{max} would not activate the FES.

The onset detection algorithm to trigger FES is divided into two sections (A and B), representing the two moments of MES acquisition in Figures 2 and 3. In A, the value for MES_1 is extracted, and a value for the threshold is built (MES_{th}). The process for section B is composed of comparisons. A second value for the window MES_2 is extracted and two comparisons are developed. If the instant MES (MES_2) value is below MES_{max} and over MES_{th} , the FES is triggered; otherwise, the algorithm loops back to the A section.

FES generation system

The developed electrical stimulation system is a biphasic wave generator that drives an amplifier. The wave has a fixed period with a variable period of active duration.

The FES intensity was adjusted by Pulse Width Modulation (PWM). The total period is $T = t_{off} + 2 * t_{on}$; t_{on} period is fixed and determined by the user, while the t_{off} is the idle period. A frequency ranging from 20 to 40 Hz with 1s of duration is executed to generate the ramp effect of FES. The diagram of Figure 3 illustrates the FES generation process.

To build a ramp with incremental frequency, the pulse period should be incremented by a value each cycle. The increment/decrement δ can be calculated using the relation:

$$\delta = \frac{T_f^2 - T_i^2}{2t_R}, \quad (4)$$

being T_f and T_i the final and initial desired pulse period, and t_R the ramp duration. For example, a ramp of 1 second, starting with 20 Hz to 40 Hz, will demand a period decrement of 937.5 μs each cycle.

The skin region was cleaned with alcohol 70° INPM. The Bracelet was positioned in the upper forearm region. More specifically, on the wrist and finger extensor muscles, ensuring that the MES and FES electrodes (rubber type) attached to the bracelet are appropriately aligned with the intended area (Botter et al., 2011). The frequency of electric stimulation was established between 20 to 40 Hz, and the pulse width was adjusted according to need.

Application to interface MES-FES

The user interface was built on four pillars: (I) easy use and visualization of its functions, (II) free access, (III) available for Android®, and (IV) Bluetooth connection to electric stimulation circuit to activate the FES, as presented on Figure 4. The application is based on a single screen that presents the control options of “difficulty,” “on time,” “intensity,” and “stop”

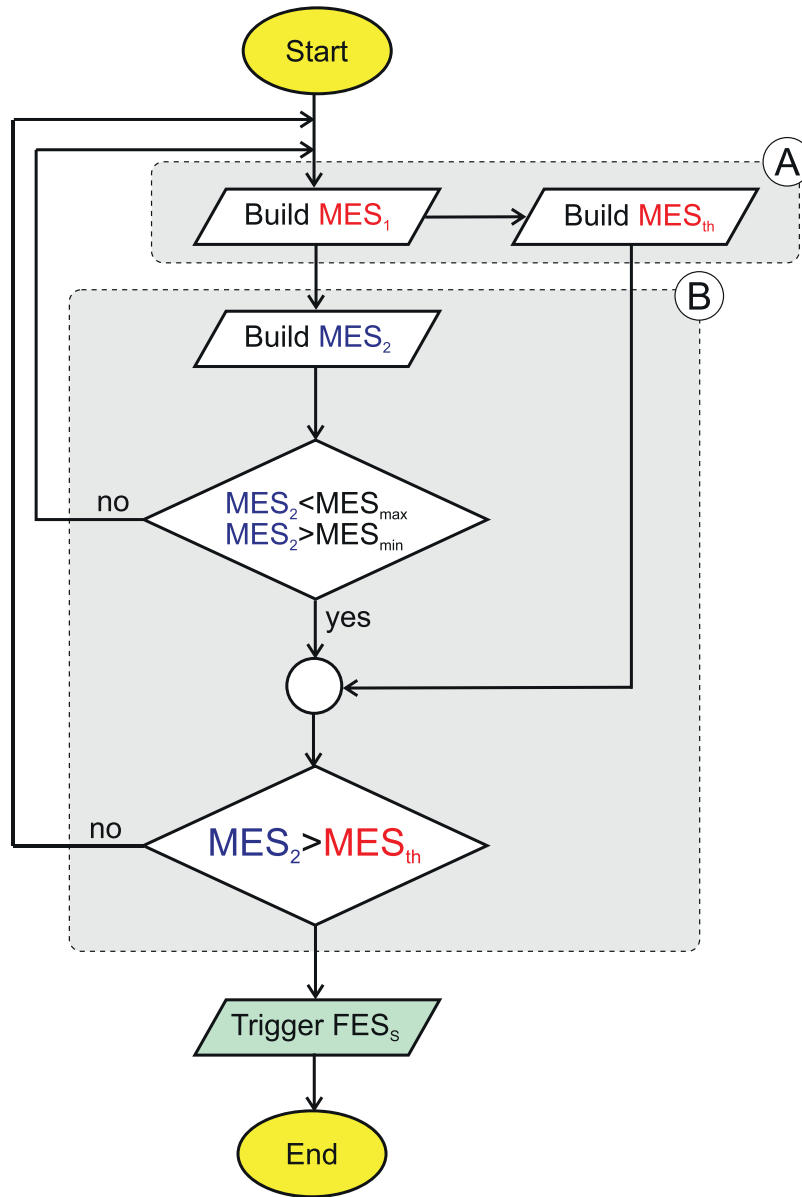


Figure 2. Strategies applied to trigger the FES circuit, which occurs by comparison of values of two instants of MES and thresholds. The routine is divided into two main parts: (A) set values for the comparison based on the signal acquisition on a time interval and (B) the comparison among the values.

functions. The **Video 1** shows the post-stroke participant where the MES_{th} was reached, although, on one side of the video, the FES has “intensity” set with “9” and the other with “0.”

The difficulty varies from 1 to 100. Increasing the difficulty will make it harder to activate the FES, i.e., a stronger muscle contraction will be necessary to start the FES. The on-time function varies between 1 s and no maximum limit; it refers to the time in seconds that the FES will remain to provide the electrical stimulus. The longer the time, the longer the stimulus maintenance. This parameter sets the ΔMES threshold and is adjusted based on MES values observed during the individual’s residual movement.

The intensity can vary from 1 to 20, and the greater this value is, the more significant the electrical stimulus will be and, consequently, a stronger muscle contraction. The intensity is based on the pulse duration modulation (t_{on}). Each step of

intensity increases $1 \mu s$ of each pulse phase. A stop function must be used in cases where it is necessary to deactivate the FES, thus reducing its intensity to 0.

Results

The final prototype weighed 133 g for the forearm bracelet and circuits (microcontroller, sEMG circuit, and Bluetooth module) and 186 g for the FES generator circuits and housing (battery charger, battery, and transformer) mounted on an armband. These aspects encompass wearability, a key criterion in the design process. As it is a lightweight, portable device without the need for wired connections, it offers greater comfort, allows greater user mobility, and therefore increases adherence, especially in-home use, as shown in the **Video 2** during a grasp event.

The peripheral electrical stimulation is a noninvasive technique; nevertheless, The FES level was adjusted to generate

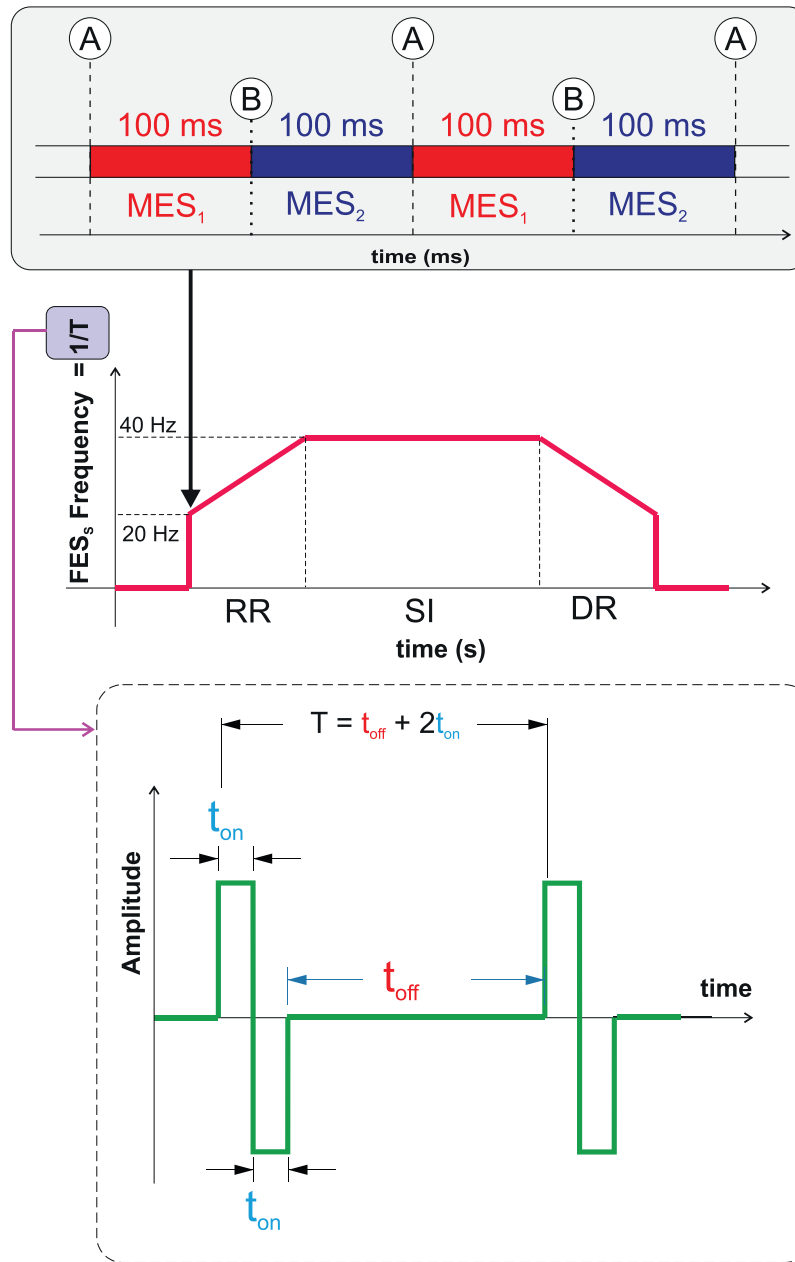


Figure 3. Method to create the electrical stimulation waveform: when the MES signal triggers the FES, a Rise Ramp (RR) starts, increasing the frequency, e.g., from 20 to 40 Hz; during the stimulation interval (SI). The frequency is sustained by a predetermined time; the decay Ramp (DR) decreases the pulse frequency.

muscle contraction while being below the pain threshold, assuring acceptance by the user. This was achieved progressively by increasing the intensity until the contraction was observed.

The MES-FES interface functions autonomously, restoring upper limb motor function in hemiplegic individuals and preserving their motricity. The wireless connection enables the configuration of device parameters using any mobile device. The configuration does not require physical access, enhancing flexibility and usability.

Among all the mentioned aspects that make neuro-orthosis accessible, the low cost of the product is paramount. The customized device in this study was constructed with low-cost materials, costing approximately 20 US\$.

A recording of the signal during the device use is shown in Figure 5. The system is voluntarily activated quickly when the

MES value surpasses the dynamic threshold, activating the FES system. Due to severe electrical interference, the MES cannot be measured during the simulation. One of the drawbacks lies in the fact that the orthoses can only be activated using MES, but not “deactivated.” Because of this, the FES time is set by the user.

An example of the FES pulse generated by the transformer (with and without load) is shown in Figure 6. In this example, the intensity was adjusted such that the pulse included a negative duration of 200 μ s and a positive duration of 200 μ s, resulting in an approximate t_{on} of 400 μ s. The pulse period of around 28 ms means a pulse frequency of 35 Hz. As the pulse is switched, only a squared pulse can be achieved. However, having an arbitrary waveform could be beneficial to avoid voltage spikes, which may cause discomfort to the

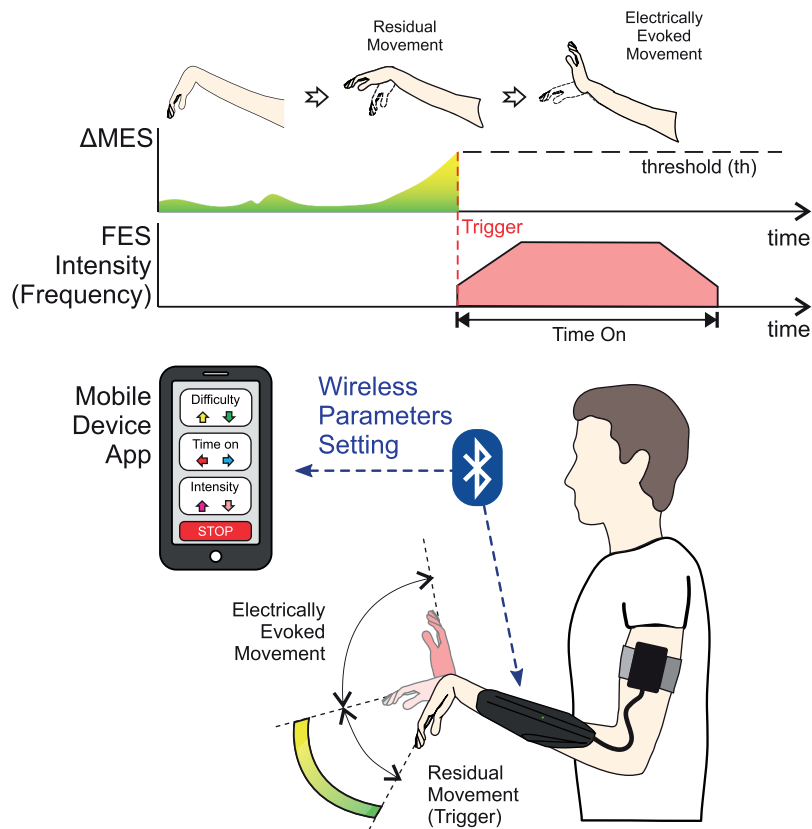


Figure 4. Illustrative non-invasive MES-FES interface application. The communication is performed by blue- tooth technology.

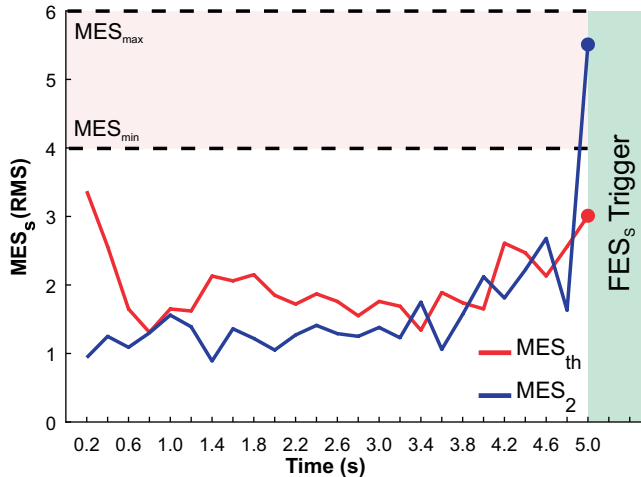


Figure 5. Post stroke participant Myoelectrical signal. FES is triggered when the MES signal surpasses the threshold while inside the acceptable values region.

user. The strategy used was having a frequency ramp, which helps avoid an abrupt input of FES.

Discussion and comparative analysis

As previously presented, the main idea of this work is to present an innovative open- source initiative: an affordable, DIY, non-intrusive, adaptable, wearable myoelectrical orthosis designed for individuals with hand disabilities. This device integrates MES and FES through a sPNSI, ensuring a noninvasive approach. By real-time detection of movement intentions

using sEMG sensors, the orthosis activates electrical stimulation to aid hand movements, enhancing users' autonomy and facilitating daily activities. Crafted with 3D additive manufacturing, the forearm-worn orthosis allows for personalized adjustments, while the stand-alone control system, featuring Bluetooth communication, ensures wireless functionality. This lightweight and customizable model, complemented by an open-source software platform, elevates usability and broadens applicability with readily available components.

Our system features a mobile device as a control interface for home-based assistive technology devices. The user can configure difficulty (threshold level of MES to trigger FES), FES intensity, and duration. Therefore, the users can adjust the suitable parameter setting to their needs. This approach can enable the use of the interface at home without needing technical assistance or daily calibration (Davis et al., 2022; Jasuja et al., 2019).

The MES-FES interface is modular and low-cost, which may ease the reproducibility of the future development of muscle interfaces. Configurable elements, such as 3D- printed parts, substantially increase flexibility, enabling users to customize features to suit their needs (Khoshmanesh et al., 2021).

As the system is wearable, it can be quickly placed on the body. To make the system wearable, we had to reduce weight and distribute the weight into separate parts based on similar works (Crepaldi et al., 2021). Making the system wearable and appropriate for in-home use is crucial as it may increase the adherence to the device use (Hung et al., 2021).

To the extent of our knowledge, a solution that integrates all these features is still non-existent in the literature. However,

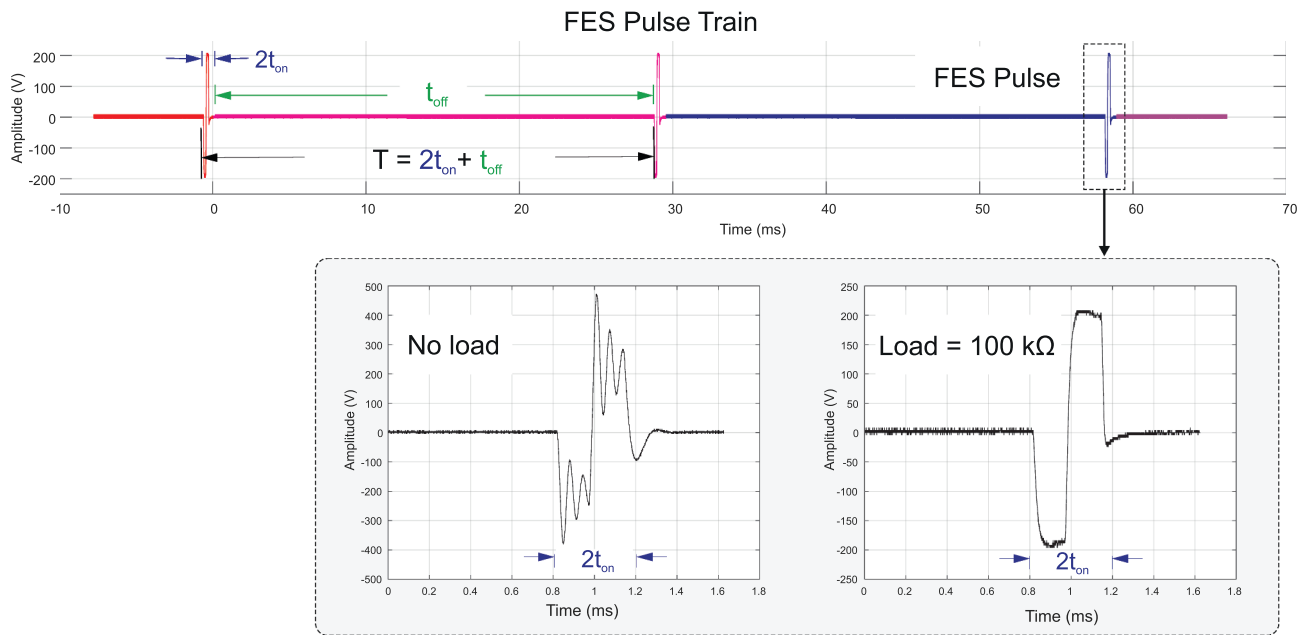


Figure 6. FES pulse generated by the system.

similar solutions deserve a more detailed comparison, as presented below.

Similar MES-FES systems have been presented in the literature, such as in (Wang, Bi, et al., 2017), where a wearable and wireless system was presented. A common element is having the onset algorithm based on a threshold adjusted by the MAV. In the present work, the threshold has a differential nature, enabling the detection of signal changes even if the baseline changes. However, some users presented a slow and gradual contraction pattern that does not easily trigger the system. Therefore, a more robust approach should be investigated among the aim population. For instance, Yu et al (Yu et al., 2002). combined MAV and gradient features as inputs for an artificial neural network (ANN) to automatically switch FES, demonstrating a potential avenue for further research.

The lack of closed-loop control is a drawback of the proposed system in this work, as a fixed time for system shut-off is set. In related work, Y. Zhou et al. (2018) proposes a closed-loop FES system that uses MES feedback from bilateral arms to enhance upper-limb stroke rehabilitation. The FES is controlled using the contralateral (unaffected) forearm as a reference. This approach may reduce unnecessary stimulation. However, when the interface is used as assistive technology, FES is applied in the same region as the MES is recorded, contaminating the signal (Zhang & Ang, 2007). Some strategies could be investigated to improve the device control, such as fusing MES with other sensors (Jović et al., 2012) and improving signal processing or instrumentation to detect movement intention even in the presence of electrical stimulation (Jung et al., 2021).

Sennels et al. proposed using volitional electromyography as a control signal for muscle stimulation and an adaptive linear prediction filter to eliminate stimulation artifacts and muscle responses. Adaptive filters have shown promising results in effectively eliminating muscle responses and artifacts from electromyographic signals. On the other hand, Increasing the filter length to achieve better

muscle response suppression can result in a longer delay, which may not be practical in real-time applications (Sennels et al., 1997).

Another strategy proposed by Widjaja et al. to reduce FES artifacts from sEMG is to use a blanking window to remove the signal portion where the stimulation artifact is likely to occur and comb filters to remove muscle responses from FES (M-waves). Many algorithms, including blanking, are deemed unable to adapt to the dynamic nature of the stimulation artifact due to the nonlinearities of the stimulation procedure and, hence, suffer residual artifacts (Widjaja et al., 2009).

The use of FES is relatively easy to implement and is non-invasive. However, it has limitations as it can only selectively activate subsets of muscles. Although electrode arrays and implantable electrodes have addressed some issues, they still face constraints, including the quick onset of muscular fatigue and limited controllability of muscular recruitment (Shokur et al., 2021). Nevertheless, including multiple MES-FES output channels is a suitable approach to augment the range of movement while keeping the system noninvasive (Wang, Guo, et al., 2017; Whicher et al., 2020). Another strategy to improve motor control is using MES as a parameter to modulate the FES (M. Chen et al., 2013; Y. X. Zhou et al., 2016).

Another limitation of the study involves the electrical transformer. Firstly, its weight (63 g) covers a large part of the total weight of the neuro-orthosis. Therefore, future studies may focus on the design of a low-weight transformer or alternatives for FES driving. Moreover, the transformer introduces distortions to the signal, which may be avoided if a DC-DC converter is employed. On the other hand, semiconductor-based circuits will not be intrinsically isolated as the transformer is.

Conclusion

This study details the design of an open-source functional electrical stimulation (FES) interface controlled by myoelectric

signals (MES) to develop a DIY myoelectric orthosis specifically targeted at surface peripheral nervous system applications in the upper limb (forearm) region. We describe the MES and electrical stimulation circuitry, software routines for FES, an Android application with Bluetooth connectivity for device control, and a bracelet model available for researchers wishing to investigate the effects of FES on forearm conditions. The device's total weight, at 326 g, is exceptionally lightweight, and its open platform design enhances wearability, avoiding the discomfort and movement artifacts often associated with cable or electrode connections.

Future investigations will focus on enhancing the application's functionality by integrating voice commands for adjusting parameters, refining the bracelet design, and exploring the feasibility of neuro-orthosis as an assistive technology or rehabilitation device. Further research will evaluate user learning effects, quantify activation errors and successes, investigate external interference, and assess user tolerance for prolonged use.

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